

[Academic Presentation Title]

[Presentation Subtitle]

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[Conference Name, Month Year]

Progressive vs. Non-progressive Visualization

Non-progressive visualization

- Renders in one pass.
- Each change in the data, whether caused by interaction or new data, triggers a full rendering.

Progressive visualization

- Rendering uses a series of passes that improve the result until completion.
- Each pass is limited to a requested latency and renders a meaningful partial result.
- Each partial result has a meaningful error or quality measure and an estimate of completion, such as a progress bar.
- Users can interact at any time; the visualization adapts as much as possible and can implement steering when needed.

Definition adapted from Ulmer et al., “A Survey on Progressive Visualization” (2024).

Progressive Computation

F_p : progressive function

q : requested interval between outputs

R_{i-1}, R_i, R : previous, partial, and final result

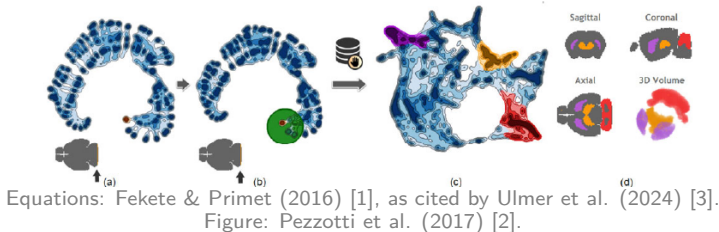
P : function parameters; D_i : data at pass i

t_i, m_i : time and memory at pass i

$R_0 = \emptyset$: first call has no previous result

$$F_p(P, D_i, q, R_{i-1}) \xrightarrow{t_i, m_i} R_i, \quad i = 1, \dots, z$$

$t_i \leq q$, $R_i \rightarrow R$ or a termination criterion is reached.



Three Ways to Produce Progressive Results



Ulmer et al. (2024), "A Survey on Progressive Visualization."

Data chunking

$$D_i = d_1 \cup \dots \cup d_i$$
$$R_i = F_p(P, D_i, q, R_{i-1})$$

R_i summarizes the data processed so far. It may be incomplete for the full dataset.

The custom-chunking formula is schematic. The survey gives no universal equation.

Process chunking

$$D_i = D$$
$$R_i = F_p(P, D, q, R_{i-1})$$

R_i is the current iteration output on the full dataset.

Custom chunking

$$D_i = \pi_i(P, D, R_{i-1})$$
$$R_i = F_p(P, D_i, q, R_{i-1})$$

R_i follows a task-specific or hybrid schedule.

The Intermediate State R_i as an Object of Analysis

$$R_0 \longrightarrow R_1 \longrightarrow \dots \longrightarrow R_i \longrightarrow \dots \longrightarrow R_z$$

The sequence $\{R_i\}$ exposes how an analysis develops over time.

Pass-indexed

Each R_i is a state at a particular pass i .

Partial

R_i is an intermediate output; it need not yet be the terminal state R_z .

Evolving

Its relation to R_{i-1} makes change across passes observable.

Candidate properties to examine: quality/error, uncertainty, change, stability, and convergence

These properties require task-specific interpretation and measures. The analytical framing is this talk's; the progressive-visualization definition is from Ulmer et al. (2024).

Progressive Visual Analytics: Two Perspectives

Ulmer et al. (2024)

“each pass limits its execution time to a requested latency and renders a meaningful partial result;”

“a progressive visualization allows interactions at any time; it adapts as much as possible, even implementing steering if necessary.”

“A Survey on Progressive Visualization” [3].

Pezzotti et al. (2017)

“Progressive Visual Analytics aims at improving the interactivity in existing analytics techniques by means of visualization as well as interaction with intermediate results.”

“With this feedback, the user can decide on local refinements and steer the approximation level during the analysis.”

“Approximated and User Steerable tSNE for Progressive Visual Analytics” [2].

Paper 1: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 1: Motivation

- *[Why the research question matters]*
- *[Limitations of existing explanations]*
- *[How the paper approaches the question]*

Paper 1: Research Questions and Hypotheses

- *[Research question]*
- *[Hypothesis or proposition 1]*
- *[Hypothesis or proposition 2, if applicable]*

Paper 1: Conceptual Framework

- *[Key concepts or variables]*
- *[Theoretical relationships and proposed mechanisms]*
- *[Insert the paper's central framework figure here]*

Paper 1: Data and Research Subjects

- *[Data or material sources and time period]*
- *[Sample composition or case selection]*
- *[Data processing and ethics, if applicable]*

Paper 1: Research Methods

- *[Research design and analytical method]*
- *[Model, experiment, or analysis workflow]*
- *[Key identification assumptions or derivation conditions]*

Paper 1: Results Overview

- *[Main result 1]*
- *[Main result 2]*
- *[How the results address the hypotheses or propositions]*

Paper 1: Main Evidence

- *[Insert the key figure, table, or theorem]*
- *[Key estimate, proof, or observation]*
- *[Conclusion supported by this evidence]*

Paper 1: Mechanisms and Additional Analyses

- *[Mechanism analysis or theoretical explanation]*
- *[Robustness, sensitivity, or alternative explanation checks]*
- *[Boundary conditions for the results]*

Paper 1: Summary

- *[Main conclusion]*
- *[Contribution and limitation]*
- *[Connection to Paper 2]*

Paper 2: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 2: Research Question and Contribution

- *[Research question]*
- *[What this paper adds to related work]*
- *[Relationship to or distinction from Paper 1]*

Paper 2: Framework and Hypotheses

- *[Theoretical basis or conceptual framework]*
- *[Hypotheses or propositions]*
- *[Proposed mechanism or reasoning]*

Paper 2: Data and Research Design

- *[Data or material sources and sample]*
- *[Research design and key variables or concepts]*
- *[Why the design fits the research question]*

Paper 2: Methods and Analysis Strategy

- *[Model, experiment, comparison, or proof method]*
- *[Identification strategy or analytical steps]*
- *[Key assumptions and how they are assessed]*

Paper 2: Main Results

- *[Main result 1 and supporting evidence]*
- *[Main result 2 and supporting evidence]*
- *[How the results answer the research question]*

Paper 2: Mechanisms and Robustness

- *[Mechanisms, heterogeneity, or boundary conditions]*
- *[Robustness or sensitivity analyses]*
- *[Alternative explanations and additional evidence]*

Paper 2: Summary and Overall Conclusions

- *[Main conclusion of Paper 2]*
- *[Shared findings and differences across the two papers]*
- *[Overall answer to the research topic]*

- [1] J.-D. Fekete and R. Primet, “Progressive Analytics: A Computation Paradigm for Exploratory Data Analysis,” CoRR, abs/1607.05162, 2016.
- [2] N. Pezzotti et al., Approximated and User Steerable t-SNE for Progressive Visual Analytics, *IEEE Transactions on Visualization and Computer Graphics*, 23(7), 1739–1752, 2017. doi:10.1109/TVCG.2016.2570755.
- [3] A. Ulmer et al., A Survey on Progressive Visualization, *IEEE Transactions on Visualization and Computer Graphics*, 30(9), 6447–6467, 2024. doi:10.1109/TVCG.2023.3346641.

[Thank You]

[Questions and Discussion]

[Presenter email, optional]